Example 1.29 Let $f, g : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = 2x - |x|$ and $g(x) = 2x + |x|$. Find $f \circ g$.

Solution:
We know

$$|x| = \begin{cases} -x & \text{if} \quad x \leq 0 \\ x & \text{if} \quad x > 0 \end{cases}$$

So

$$f(x) = \begin{cases} 2x - (-x) & \text{if} \quad x \leq 0 \\ 2x - x & \text{if} \quad x > 0 \end{cases}$$

Thus

$$f(x) = \begin{cases} 3x & \text{if} \quad x \leq 0 \\ x & \text{if} \quad x > 0 \end{cases}$$

Also

$$g(x) = \begin{cases} 2x + (-x) & \text{if} \quad x \leq 0 \\ 2x + x & \text{if} \quad x > 0 \end{cases}$$

Thus

$$g(x) = \begin{cases} x & \text{if} \quad x \leq 0 \\ 3x & \text{if} \quad x > 0 \end{cases}$$

Let $x \leq 0$. Then

$$(f \circ g)(x) = f(g(x)) = f(x) = 3x.$$

The last equality is taken because $3x \leq 0$ whenever $x \leq 0$.

Let $x > 0$. Then

$$(f \circ g)(x) = f(g(x)) = f(3x) = 3x.$$

Thus $(f \circ g)(x) = 3x$ for all x.

1.6.5 Inverse of a Function

Let there be a bijection $f : X \to Y$ as given in the Figure 1.41.

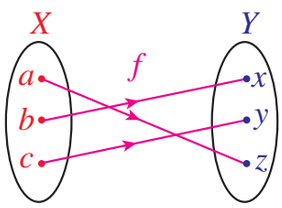


Figure 1.41

If we look this function in a mirror, we get a function from Y to X. Let us call that function as g. Then g is a function from Y to X defined by $g(x) = b, g(y) = c, g(z) = a$.

This function g is an example for the inverse of f. Now we define the inverse of a function.